

3 INTEGER & MATRIX MULTIPLICATION

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1: procedure KMULT(x, y, n)
2:                                     ▷ x, y are positive n-bit numbers
3:   if n = 1 then return x · y
4:   else
5:     k ← ⌈ $\frac{n}{2}$ ⌋
6:     a ← x[1 .. ⌊ $\frac{n}{2}$ ⌋], b ← x[⌊ $\frac{n}{2}$ ⌋ + 1 .. n]
7:     c ← y[1 .. ⌊ $\frac{n}{2}$ ⌋], d ← y[⌊ $\frac{n}{2}$ ⌋ + 1 .. n]
8:     vac ← KMULT(a, c, k), vbd ← KMULT(b, d, k)
9:     w ← KMULT(|a − b|, |c − d|, k)
10:    t ← vac + vbd − sign(a − b) · sign(c − d) · w
11:    return  $2^{2k} v_{ac} + 2^k t + v_{bd}$ 

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Recurrence: $T[n] \leq 3 \cdot T[\lceil \frac{n}{2} \rceil] + c \cdot n$

$$\Rightarrow T[n] = O(n^{\log_2(3)}) = O(n^{1.585})$$

The numbers *x, y* are non-negative, and thus also *a, b, c, d*, and thus also *v_{ac}, v_{bd}* are all non-negative. But (*a* − *b*) and (*c* − *d*), and therefore also *w*, might be negative. Therefore we compute the sign of *w* in Line 10. We let the sign function for any number *z* be defined as

$$\text{sign}(z) = \begin{cases} +1 & \text{if } z \geq 0 \\ -1 & \text{if } z < 0 \end{cases}$$

The value *t* is always non-negative: It would be negative if $ac + bd < (a - b)(c - d)$, but $(a - b)(c - d) = ac - ad - bc + bd$, and $ac + bd < ac - ad - bc + bd$ is equivalent to $ad + bc < 0$, which can't happen because *a, b, c, d* are all non-negative.

For determining the runtime, apply the recursion tree method:

Level		# Calls	Merge $c \cdot n$	Total runtime
0	N	1	$c \cdot N$	$1 \cdot c \cdot N$
1	$\frac{N}{2} \dots \frac{N}{2}$	3	$c \cdot \frac{N}{2}$	$3 \cdot c \cdot \frac{N}{2}$
2	$\frac{N}{4} \dots \frac{N}{4} \dots \frac{N}{4} \dots \frac{N}{4}$	9	$c \cdot \frac{N}{4}$	$9 \cdot c \cdot \frac{N}{4}$
d	$\frac{N}{2^d} \dots \frac{N}{2^d} \dots \frac{N}{2^d} \dots \frac{N}{2^d}$	3^d	$c \cdot \frac{N}{2^d}$	$3^d \cdot c \cdot \frac{N}{2^d}$
$\lceil \log(N) \rceil$	1 1 1 1 1 1 1 1	N	c	$c \cdot N$

$$\sum_{d=0}^{\lceil \log(N) \rceil} 3^d \cdot c \cdot \frac{N}{2^d}$$

Solving the sum with the partial sum formula:

$$\sum_{d=0}^{\lceil \log(N) \rceil} 3^d \cdot c \cdot \frac{N}{2^d} = c \cdot N \cdot \sum_{d=0}^{\lceil \log(N) \rceil} \frac{3^d}{2^d} = c \cdot N \cdot \frac{\frac{3}{2}^{\lceil \log(N) \rceil + 1} - 1}{\frac{3}{2} - 1} = O(N \cdot \frac{3}{2}^{\log(N)}) = O(N^{1 + \log(\frac{3}{2})}) = O(N^{\log(3)})$$